KL Mirror Descent

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$$\min_{x \in \Delta} f(x) \qquad (f: \text{convex}) \qquad \text{(1)}$$

where Δ is the probability simplex, i.e.,

$$\Delta = \left\{ x : \sum_{i=1}^{n} x(i) = 1, \; x(i) \geq 0 \right\}$$

By analogy to the proximal interpretation of gradient but with a KL regularizer, consider

$$x_{k+1} \in \arg\min_{x \in \Delta} f(x_k) + \nabla f(x_k)(x - x_k) + \frac{1}{\varepsilon_k} D_{KL}(x \| x_k) \qquad \text{(A)}$$

where $D_{KL}(x \| x_k) = \sum_i x(i) \log \frac{x(i)}{x_k(i)}$

Differentiate w.r.t. $x(i)$

$$\nabla f(x_k)(i) + \frac{1}{\varepsilon_k} \left(\log \frac{x(i)}{x_k(i)} + x(i) \frac{1}{\frac{x(i)}{x_k(i)}} \cdot \frac{1}{x_k(i)} \right) = 0$$

$$\log \frac{x(i)}{x_k(i)} = -1 - \varepsilon_k \nabla f(x_k)(i)$$

$$\Rightarrow \quad x_{k+1}(i) = \frac{x_k(i)\, e^{-\varepsilon_k \nabla f(x_k)(i)}}{\sum_j x_k(j)\, e^{-\varepsilon_k \nabla f(x_k)(j)}}$$

normalizing

$$x_{k+1}(j) = \frac{x_k(j)\, e^{-\varepsilon_k \nabla f(x_k)(j)}}{\sum_j x_k(j)\, e^{-\varepsilon_k \nabla f(x_k)(j)}}$$

normalizing to make it a probability distribution

(can be made rigorous using Lagrange multiplier theory)

We want to prove that this algorithm converges i.e., $x_k \to x^*$ where x^* solves (1)

To prove the result, we will use the following facts:

(1) $D_{KL}(p\|q) \geq \frac{1}{2}\|p-q\|_1^2$ (Pinsker's inequality)

(2) If g is a convex function over a convex set C and y^* is its minimum, then

$$\nabla^T g(y^*)(y-y^*) \geq 0 \quad \forall\, y \in C$$

We will apply (2) to problem (A)

$$\min_{x \in \Delta} \nabla^T f(x_k)\, x + \frac{1}{\varepsilon_k} D_{KL}(x\|x_k)$$

$D_{KL}(x\|x_k)$: can show this is convex in x.

Since x_{k+1} is the optimal solution

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$$\left(\nabla f(x_k) + \frac{1}{\varepsilon_k} \nabla_x D_{KL}(x_{k+1} \| x_k)\right)^T (x - x_{k+1}) \geq 0 \quad \forall x$$

$$\nabla_x D_{KL}(x \| x_k)(i) = \frac{\partial}{\partial x(i)} \sum_i x(i) \log \frac{x(i)}{x_k(i)}$$

$$= \log \frac{x(i)}{x_k(i)} +$$

$$\Rightarrow \quad \nabla_x D_{KL}(x_{k+1} \| x_k)(i) = \log \frac{x_{k+1}(i)}{x_k(i)} + 1$$

$$= \log x_{k+1}(i) + 1 - \log x_k(i)$$

$$\Rightarrow \quad \nabla_x D_{KL}(x_{k+1} \| x_k) = \nabla h(x_{k+1}) - \nabla h(x_k) + \vec{1}$$

where $\quad h(x) = \sum_i x_i \log x_i$

$$\Rightarrow \quad \left(\nabla f(x_k) + \frac{1}{\varepsilon_k}\left(\nabla h(x_{k+1}) - \nabla h(x_k) + \vec{1}\right)\right)^T (x^* - x_{k+1}) \geq 0$$

$$1^T (x^* - x_{k+1}) = 1 - 1 \geq 0$$

$$\Rightarrow \quad \left(\nabla f(x_k) + \frac{1}{\varepsilon_k}\left(\nabla h(x_{k+1}) - \nabla h(x_k)\right)\right)^T (x^* - x_{k+1}) \geq 0$$

Now, consider the Lyapunov function : $D(x^* \| x_n)$

$$D(x^* \| x_{n+1}) - D(x^* \| x_n) = \sum_i \left(\log \frac{x^*(i)}{x_{n+1}(i)} - \log \frac{x^*(i)}{x_n(i)} \right) x^*(i)$$

$$= \sum_i x^*(i) \left(\log x_n(i) - \log x_{n+1}(i) \right)$$

$$= x^{*T} \left(\nabla h(x_n) - \nabla h(x_{n+1}) \right)$$

$$= \left(x^* - x_{n+1} \right)^T \left(\nabla h(x_n) - \nabla h(x_{n+1}) \right)$$

$$+ x_{n+1}^T \left(\nabla h(x_n) - \nabla h(x_{n+1}) \right)$$

$$\leq \varepsilon_n \nabla^T f(x_n) \left(x^* - x_{n+1} \right) + \sum_i x_{n+1}(i) \log \frac{x_n(i)}{x_{n+1}(i)}$$

$$= \varepsilon_n \nabla^T f(x_n) \left(x^* - x_{n+1} \right) - D_{KL}\left(x_{n+1} \| x_n \right)$$

$$\leq \varepsilon_n \nabla^T f(x_n) \left(x^* - x_n \right) + \varepsilon_n \nabla^T f(x_n) \left(x_n - x_{n+1} \right)$$

$$- D_{KL}\left(x_{n+1} \| x_n \right)$$

Holder's

$$\leq \varepsilon_n \left(f(x^*) - f(x_n) \right) + \varepsilon_n \| \nabla f(x_n) \|_\infty \| x_n - x_{n+1} \|_1$$

$$- D_{KL}\left(x_{n+1} \| x_n \right)$$

$$f(x^*) \geq f(x_k) + \nabla^T f(x_k)(x^* - x_k) \qquad - D_{KL}(x_{k+1} \| x_k)$$

(convexity)

$$\leq \varepsilon_k \left(f(x^*) - f(x_k)\right) + \frac{1}{2}\left(\varepsilon_k^2 \|\nabla f(x_k)\|_\infty^2 + \|x_k - x_{k+1}\|_1^2\right)$$

$$ab \leq \frac{(a^2 + b^2)}{2}$$

$$- D_{KL}(x_{k+1} \| x_k)$$

$$\leq \varepsilon_k \left(f(x^*) - f(x_k)\right) + \frac{\varepsilon_k^2}{2} G_1^2, \quad \text{where}$$

$$G_1^2 \triangleq \max_x \|\nabla f(x)\|_\infty$$

(Pinsker)

$$\Rightarrow \quad f(x_k) - f(x^*) \leq \frac{1}{\varepsilon_k}\left(D_{KL}(x^* \| x_k) - D_{KL}(x^* \| x_{k+1})\right) + \frac{G_1^2}{2}\varepsilon_k$$

Let $\varepsilon_k = \varepsilon$. Telescoping and using $D_{KL}(x^* \| x_T) \geq 0$,

$$\sum_{k=0}^{T-1}\left(f(x_k) - f(x^*)\right) \leq \frac{1}{\varepsilon} D_{KL}(x^* \| x_0) + \varepsilon \frac{G_1^2}{2} T$$

$$\frac{1}{T}\sum_{k=0}^{T-1}\left(f(x_k) - f(x^*)\right) \leq \frac{1}{\varepsilon T} D_{KL}(x^* \| x_0) + \varepsilon \frac{G_1^2}{2}$$

$$\frac{1}{T}\sum_{t=0}^{T-1}\left(f(x_t)-f(x^*)\right) \le \frac{1}{\varepsilon T} D_{KL}(x^* \| x_0) + \frac{\varepsilon G^2}{2}$$

choose x_0 to be uniform distribution.

$$\Rightarrow D_{KL}(x^* \| x_0) = \sum_i x^*(i) \log n x^*(i)$$

$$= \sum_i x^*(i) \log x^*(i) + \log n$$

$$\le \log n \qquad (\text{since } \log x^*(i) \le 0)$$

$$\Rightarrow \frac{1}{T}\sum_{t=0}^{T-1}\left(f(x_t)-f(x^*)\right) \le \frac{\log n}{\varepsilon T} + \frac{\varepsilon G^2}{2}$$

choose ε: $\frac{\log n}{\varepsilon T} = \frac{G^2}{2}\varepsilon \Rightarrow \varepsilon^2 = \frac{2\log n}{G^2 T}$

$$\Rightarrow \varepsilon = \frac{1}{G}\sqrt{\frac{2\log n}{T}}$$

$$\Rightarrow \frac{\log n}{\varepsilon T} = \frac{\log n}{\frac{1}{G}\sqrt{2\log n}\sqrt{T}} = G\sqrt{\frac{\log n}{2T}}$$

$$\Rightarrow \quad \frac{1}{T}\sum_{t=0}^{T-1}\left(f(x_t) - f(x^*)\right) \le O\left(\frac{1}{\sqrt{T}}\right)$$

By convexity $\quad f\left(\frac{1}{T}\sum_{t=0}^{T-1} x_t\right) \le \frac{1}{T}\sum_{t=0}^{T-1} f(x_t)$

$$\Rightarrow \quad \boxed{f\left(\frac{1}{T}\sum_{t=0}^{T-1} x_t\right) - f(x^*) = O\left(\frac{1}{\sqrt{T}}\right)}$$

Online mirror descent:

- A sequence of functions $f_0, f_1, f_2, \ldots$
- f_t is revealed after x_t is chosen

Goal: minimize regret $R_T = \sum_{t=0}^{T-1} f_t(x_t) - \sum_{t=0}^{T-1} f_t(x^*)$

where $x^* \in \arg\min \sum_{t=0}^{T-1} f_t(x)$ (best in hindsight)

Use the same algorithm and analysis as before:

$$f(x_t) - f(x^*) \le \frac{1}{\eta_t}\left(D_{KL}(x^* \| x_t) - D_{KL}(x^* \| x_{t+1})\right)$$

$$+ \frac{\eta_t^2}{2}\varepsilon_t$$

$$+ \frac{G^2}{2}\varepsilon_k$$

$$\Rightarrow \sum_{t=0}^{T-1}\left(f_t(x_t) - f_t(x^*)\right) \le \frac{\log n}{\varepsilon} + \frac{G\varepsilon T}{2}$$

Choosing ε as before: $\boxed{R_T = O(\sqrt{T})}$